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Unit 1 Exam Scratch Paper

18)

$$\boxed{L = \int \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx}$$

$y = \frac{1}{6}x^3 + \frac{1}{2}x$ $3x^2(\frac{1}{6})$

$y' = \frac{1}{2}x^2 + \frac{1}{2}$

$$L = \int_1^3 \sqrt{1+\left[\frac{1}{2}x^2+\frac{1}{2}\right]^2}\,dx$$

$(y')^2 = (\frac{1}{2}x^2+\frac{1}{2})(\frac{1}{2}x^2+\frac{1}{2})$

$(y')^2 = \frac{1}{4}x^4 + \frac{1}{4}x^2 + \frac{1}{4}x^2 + \frac{1}{4}$

$$L = \int_1^3 \sqrt{1 + \frac{1}{4}x^4 + \frac{1}{2}x^2 + \frac{1}{4}}\,dx$$

$(y')^2 = \frac{1}{4}x^4 + \frac{1}{2}x^2 + \frac{1}{4}$

$$L = \int_1^3 \sqrt{\frac{5}{4} + \frac{1}{4}x^4 + \frac{1}{2}x^2}\,dx$$

$$L = \int_1^3$$

8) $V = \pi\int R^2 - r^2$

$$V = \pi\int_0^8 (x+6)^2 - (x^3)^2\,dx$$

1) $y = \sqrt{9-x^2}$

$$V = \pi\int_0^3 (\sqrt{9-x^2})^2\,dx$$

$$= \pi\int_0^3 (9-x^2)\,dx$$

$$= \pi\left(9x - \frac{x^3}{3}\right)\Big|_0^3 = \pi[27-9] = 18\pi$$

2) $V = 2\pi\int rh$

$$2\pi\int_0^8 x\left(x+6-x^3\right)dx$$

$$2\pi\int_0^8 x^2+6x-x^4\,dx$$

11) $x=0$ $x=\pi$

$f(x) = \sin x + 2$

$W = (\int_0^\pi \sin x + 2)$

$-\cos x + 2x\big|_0^\pi$

$(1+2\pi) - (-1) = 1+2\pi+1$

$\boxed{2\pi+2}$

x-axis

64 +

(-1, 0)

3) $\int 2\pi rh$

$$2\pi\int_0^4 (y)(-y^2+5y)\,dy$$

$$2\pi\int_0^4 -y^3+5y^2\,dy$$

volume

12) $x = \frac{4}{y}$, $x=1$, $y=1$, $y=4$ y-axis

Washer

$$V = \pi\int_1^4 \left(\frac{4}{y}\right)^2 - 1^2\,dy$$

$$= \pi\int_1^4 \frac{16}{y^2} - 1\,dy$$

$16y^{-2}$

$$= \pi(-16y^{-1} - y)\Big|_1^4$$

$$= \pi\left(\frac{-16}{4} - 4\right) - (-16-1)$$

$\pi(-4-4+17)$

$\pi(-11)$

5) $F = kx$

$20 = 0.2k$

$k = 100$

6) $W = Fd$, $F = kx$

$0.4(100) = 40$

7) $-0.1(100) = -10N$

13) $y = \frac{3}{4}x \rightarrow$ arc length $[0,8]$

$$L = \int \sqrt{1 + \left(\frac{3}{4}\right)^2}\, dx$$

$$L = \int \sqrt{1 + \frac{9}{16}} = \int_0^8 \sqrt{\frac{25}{16}} = \int_0^8 \frac{5}{4} = \frac{5}{4}x \Big|_0^8 = 10$$

14) $m = \int_0^3 \left(x + \frac{e^x}{3}\right) dx$

$$m = \left[\frac{x^2}{2} + \frac{e^x}{3}\right]\Big|_0^3$$

$$\frac{9}{2} + \frac{e^3}{3} - 0 - \frac{1}{3}$$

?

16) $y = \sqrt{x}$ $y = 0$ $x = 4$ $y = -1$

$$V = \pi \int_0^1 (\sqrt{x} + 1)^2 - 1^2 \, dx$$

$$\pi \left(x + 2\sqrt{x} + 1 - 1\right)\Big|_0^1$$

17) $A = 60\,dy$ $dF = 1000(9.8)(60)$

$$W = \int_0^2 dF\,(3-y)\,dy$$

$$W = \int (\rho g \cdot \text{area} \cdot \text{distance})\, dy$$

18) front side